n	3	4	5	6	7	8	9	10
F_n	2	3	5	8	13	21	34	55

(i)

$$F_i = F_{i-1} + F_{i-2} < 2F_{i-1}$$

$$\therefore \frac{1}{F_i} > \frac{1}{2F_{i-1}}$$

For $i \geq 4$,

$$\frac{1}{F_i} > \frac{1}{2F_{i-1}}$$

$$\frac{1}{F_4} > \frac{1}{2F_3}, \quad \frac{1}{F_5} > \frac{1}{2^2F_3}, \quad \frac{1}{F_6} > \frac{1}{2^3F_3}, \ \cdots$$

$$S > 1 + 1 + \frac{1}{2} + \frac{1}{2F_3}\left(1 + \frac{1}{2} + \frac{1}{4} + \cdots\right) = 3$$

For $i \geq 3$,

$$F_{i+2} = F_{i+1} + F_i > 2F_i$$

$$\frac{1}{F_{i+2}} < \frac{1}{2F_i}$$

$$S = 1 + 1 + \left(\frac{1}{F_3} + \frac{1}{F_5} + \cdots\right) + \left(\frac{1}{F_4} + \frac{1}{F_6} + \cdots\right)$$
$$S < 1 + 1 + \frac{1}{F_3}\left(1 + \frac{1}{2} + \frac{1}{4} + \cdots\right) + \frac{1}{F_4}\left(1 + \frac{1}{2} + \frac{1}{4} + \cdots\right)$$
$$S < 3\frac{2}{3}$$

(ii)

$$S > 1 + 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{5} + \frac{1}{8} + \frac{1}{13} = \frac{5047}{1560} > 3.2$$

$$S = 1 + 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{5} + \left(\frac{1}{F_6} + \frac{1}{F_8} + \cdots\right) + \left(\frac{1}{F_7} + \frac{1}{F_9} + \cdots\right)$$
$$< \frac{91}{30} + \frac{2}{F_6} + \frac{2}{F_7} \approx 3.03 + 0.24 + 0.15 = 3.42$$

$$\because 3.42 < 3.5$$

$$\therefore S < 3.5$$

This solution was submitted by **Jon**

Published and refined by **Jon**

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